

BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE PROJECT

SUBMITTED TO
BLUESTOCK FINTECH

SUBMITTED BY
AKSHAT BANSAL

INTERNSHIP PROGRAM
BLUESTOCK MUTUAL FUND ANALYTICS
INTERNSHIP

Technologies Used

Python ▪ Pandas ▪ NumPy ▪ SQLite ▪ Streamlit ▪ Plotly ▪ Git

Submission Date

12th June 2026

Acknowledgement

I would like to express my sincere gratitude to **Bluestock Fintech** for providing me with the opportunity to work on this Mutual Fund Analytics Capstone Project.

This internship allowed me to apply data engineering, data analytics, visualization, and software development concepts in a real-world financial domain. Through this project, I gained practical experience in data cleaning, exploratory analysis, advanced analytics, dashboard development, and project deployment.

I am thankful to the Bluestock team for their guidance and support throughout the internship.

Executive Summary

The Mutual Fund Analytics Capstone Project was developed to analyze various aspects of the Indian mutual fund industry using a comprehensive dataset consisting of fund performance, portfolio holdings, investor transactions, SIP inflows, benchmark indices, and industry-level statistics.

The project follows a complete data analytics lifecycle:

- ❖ Data Ingestion
- ❖ Data Cleaning and Transformation
- ❖ Exploratory Data Analysis
- ❖ Performance Analytics
- ❖ Dashboard Development
- ❖ Advanced Analytics
- ❖ Reporting and Deployment

A Streamlit-based interactive dashboard was developed to enable users to explore mutual fund performance, investor behavior, market trends, and risk metrics.

Advanced analytics techniques such as Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Cohort Analysis, SIP Continuity Analysis, and Herfindahl-Hirschman Index (HHI) were implemented to provide deeper insights.

The project demonstrates how data-driven decision-making can assist investors and fund managers in evaluating risks, identifying opportunities, and monitoring industry trends.

Project Objectives

- The major objectives of this project were:
- Build an end-to-end mutual fund analytics solution.
- Design an ETL pipeline for data processing.
- Perform Exploratory Data Analysis (EDA).
- Evaluate fund performance using risk-adjusted metrics.
- Analyze investor transaction behavior.
- Develop an interactive dashboard.
- Implement advanced financial analytics.
- Generate actionable investment insights.

Data Sources

The project utilizes multiple datasets stored in SQLite database tables.

Table Name	Description
fact_fund_master	Fund metadata (ID, name, category, fund house, launch date, AUM, risk level, fund manager, benchmark ID)
fact_performance	Fund performance metrics (returns for 1m/3m/6m/1y/3y/5y, Sharpe ratio, Sortino, alpha, beta, standard deviation)
fact_nav	Net Asset Value data (date, NAV value, repurchase price, sale price)
fact_transactions	Investor transaction history (investor ID, fund ID, date, type - purchase/redemption/switch, units, NAV, amount)
fact_aum_house	Fund house AUM information (fund house, date, total AUM, equity AUM, debt AUM, hybrid AUM)
fact_monthly_sip_inflows	SIP inflow trends (fund ID, month, SIP amount in Cr, new SIP count, closed SIP count)
fact_category_inflows	Category-wise inflows (category name, month, net inflow Cr, gross purchase, gross redemption)
fact_industry_folio_count	Industry folio statistics (industry name, date, folio count, total AUM Cr)
fact_portfolio_holdings	Portfolio holdings (fund ID, date, security name, ISIN, sector, holding type, market value Cr, % of AUM, quantity)
fact_benchmark_indices	Benchmark market data (index name, date, closing value, open, high, low, volume, 1-day return)

EXPLORER

BLUESTOCK_MF_CAPSTON

.streamlit

.vscode

dashboard

data

db

bluestock_mf.db

processed

raw

notebooks

reports

scripts

sql

venv

.gitignore

README.md

requirements.txt

bluestock_mf.db

data > db > bluestock_mf.db

Filter 10 tables...

TABLES

fact_aum_house

fact_benchmark_indices

fact_category_inflows

fact_fund_master

fact_industry_folio_count

fact_monthly_sip_inflows

fact_nav

fact_performance

fact_portfolio_holdings

fact_transactions

Upgrade to PRO

SELECTION

TABLE INFO

No row selected

Filter 90 rows...

num_sch...#

Filter

199

aum_crore#

Filter

278000

aum_lak...#

Filter

2.78

fund_house

Filter...

Aditya Birla Sun Life MF

abs

date

Filter

2022-03-31

1

2022-03-31

Aditya Birla Sun Life MF

2.78

278000

199

2

2022-09-30

Aditya Birla Sun Life MF

2.85

285000

199

3

2023-03-31

Aditya Birla Sun Life MF

2.75

275000

199

4

2023-09-30

Aditya Birla Sun Life MF

3.08

308000

199

5

2024-03-31

Aditya Birla Sun Life MF

3.4

340000

199

6

2024-09-30

Aditya Birla Sun Life MF

3.62

362000

199

7

2024-12-31

Aditya Birla Sun Life MF

3.84

384000

199

8

2025-03-31

Aditya Birla Sun Life MF

3.85

385000

199

9

2025-12-31

Aditya Birla Sun Life MF

4.6

460000

199

10

2022-03-31

Axis Mutual Fund

2.5

250000

95

11

2022-09-30

Axis Mutual Fund

2.4

240000

95

12

2023-03-31

Axis Mutual Fund

2.41

241000

95

13

2023-09-30

Axis Mutual Fund

2.6

260000

95

14

2024-03-31

Axis Mutual Fund

2.8

280000

95

15

2024-09-30

Axis Mutual Fund

2.9

290000

95

16

2024-12-31

Axis Mutual Fund

3

300000

95

17

2025-03-31

Axis Mutual Fund

3.1

310000

95

18

2025-12-31

Axis Mutual Fund

3.5

350000

95

19

2022-03-31

DSP Mutual Fund

1.1

110000

88

20

2022-09-30

DSP Mutual Fund

1.12

112000

88

21

2023-03-31

DSP Mutual Fund

1.15

115000

88

22

2023-09-30

DSP Mutual Fund

1.32

132000

88

23

2024-03-31

DSP Mutual Fund

1.55

155000

88

24

2024-09-30

DSP Mutual Fund

1.72

172000

88

25

2024-12-31

DSP Mutual Fund

1.88

188000

88

26

2025-03-31

DSP Mutual Fund

1.95

195000

88

27

2025-12-31

DSP Mutual Fund

2.3

230000

88

28

2022-03-31

HDFC Mutual Fund

4.35

435000

195

ETL Pipeline

The project follows a structured ETL workflow.

Extract

Raw CSV files were collected and loaded into Python.

Transform

Transformations included:

Missing value handling

Date formatting

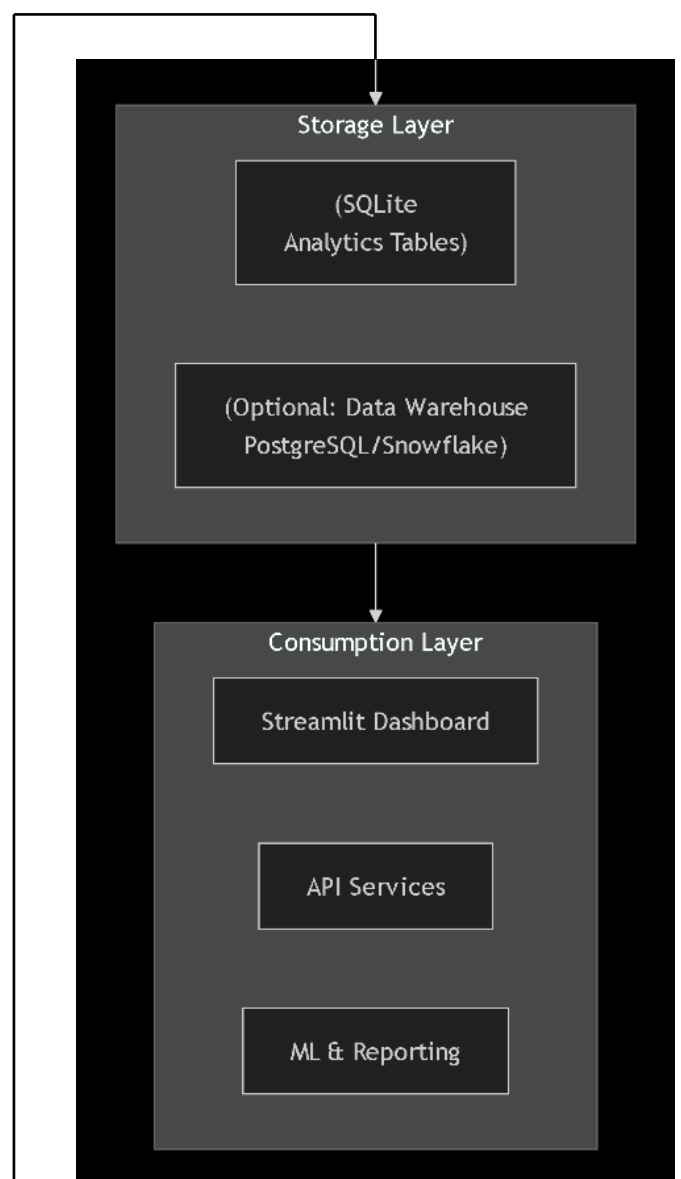
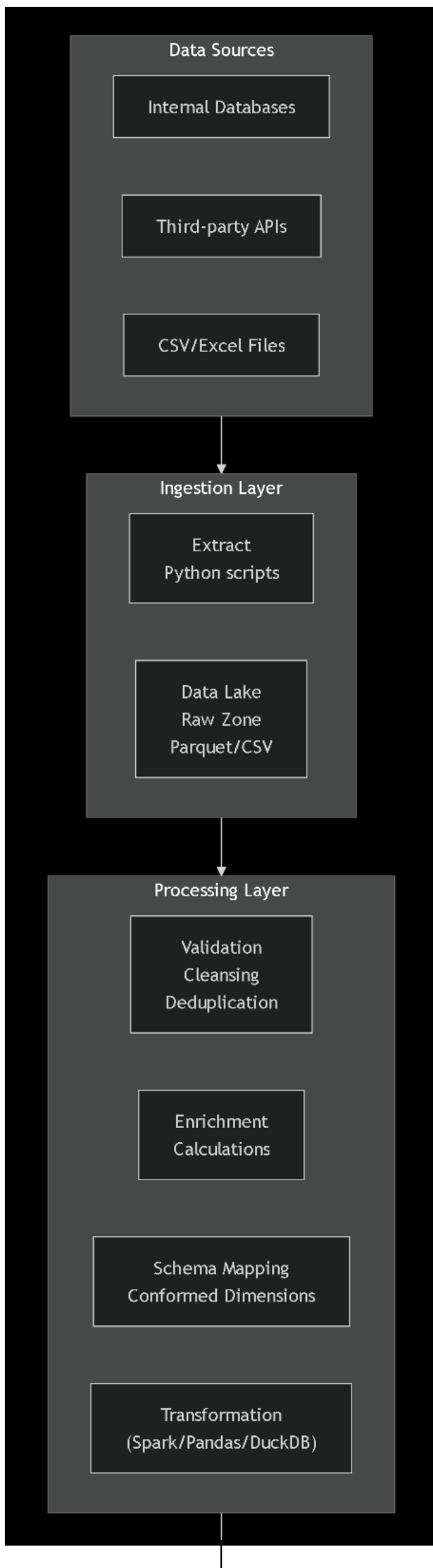
Feature engineering

Data validation

Type conversions

Load

Processed datasets were loaded into SQLite database tables.



Data Cleaning Process

Several preprocessing techniques were applied:

Missing Values

Missing values were identified and treated using appropriate strategies.

Duplicate Removal

Duplicate records were removed to ensure data quality.

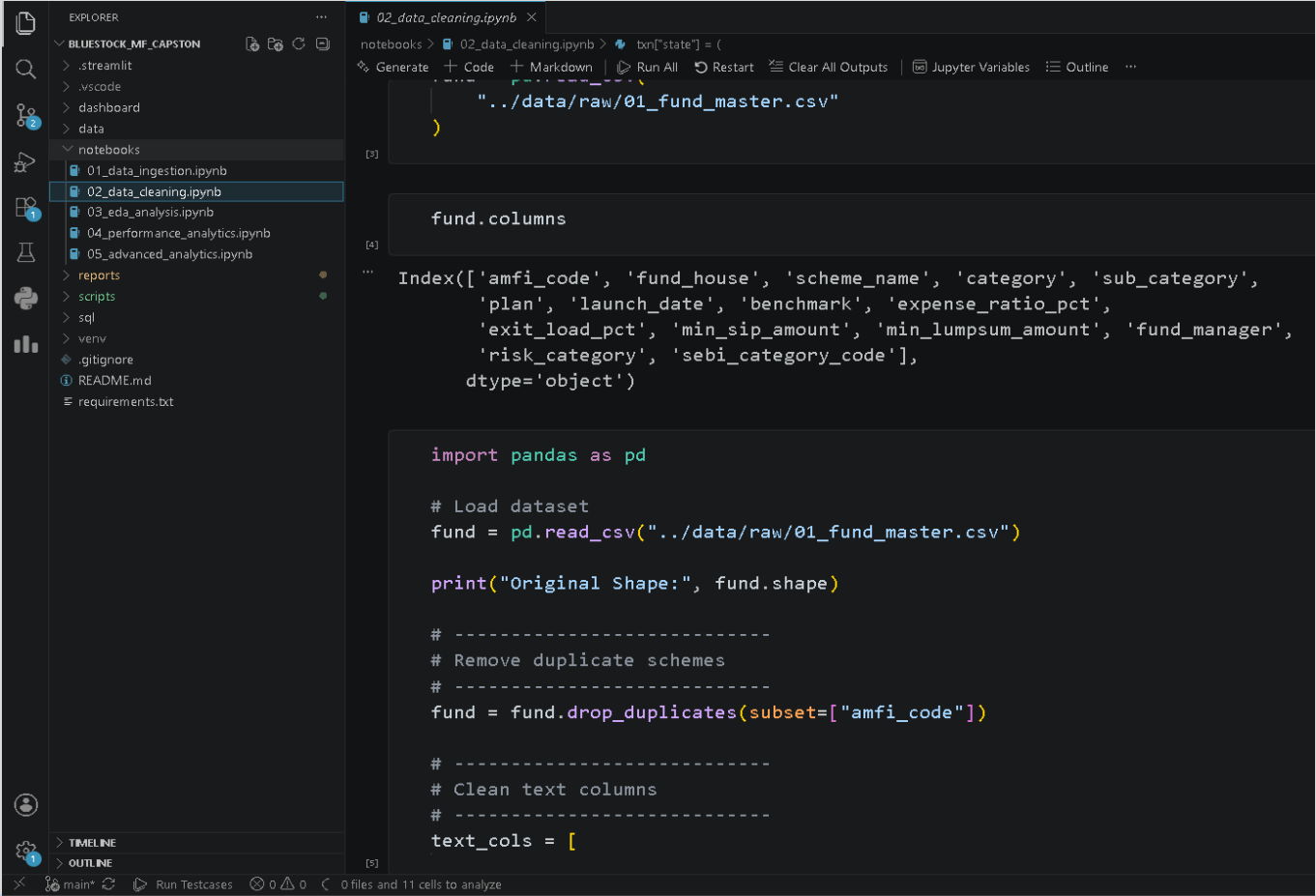
Date Processing

All date fields were standardized.

Feature Engineering

New features created include:

- CAGR
- Sharpe Ratio
- Alpha
- Beta
- Sortino Ratio
- Recommendation Score



EXPLORATORY DATA ANALYSIS

Industry Analysis

Assets Under Management (AUM)

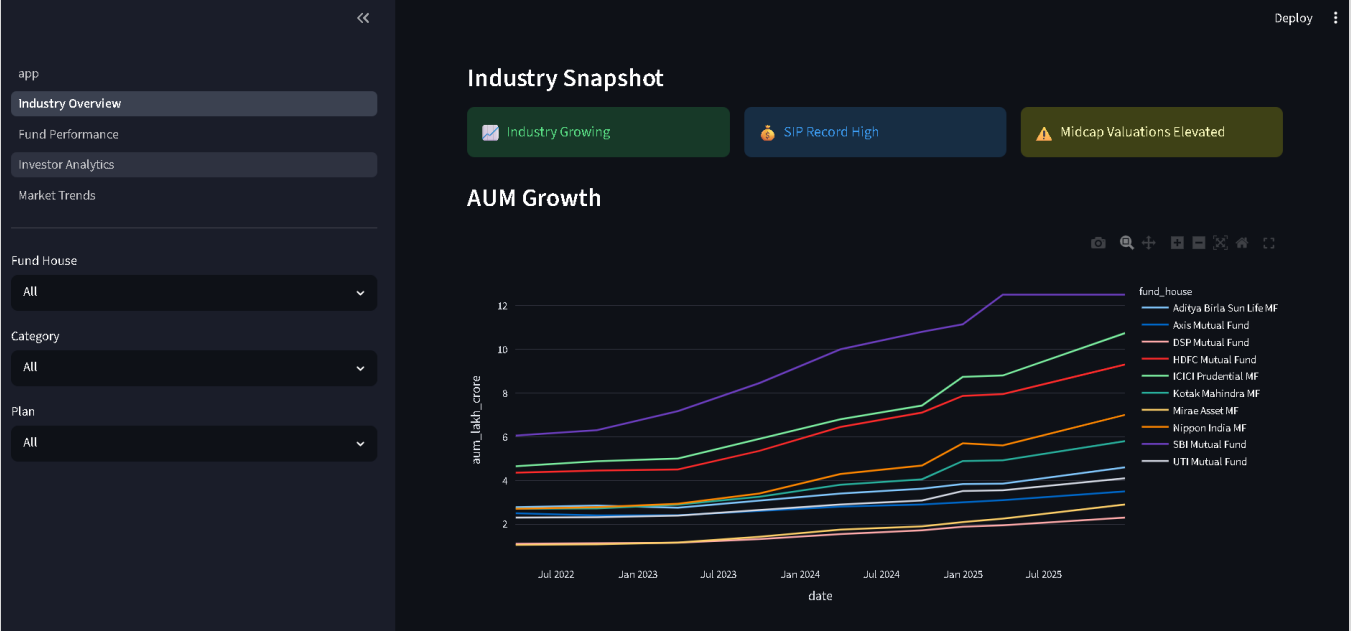
Industry AUM showed significant growth over the analysis period.

SIP Inflows

Monthly SIP inflows displayed consistent upward trends.

Folio Growth

Investor participation increased steadily.



Fund Performance Analysis

Key performance indicators analyzed:

Returns

Sharpe Ratio

Alpha

Beta

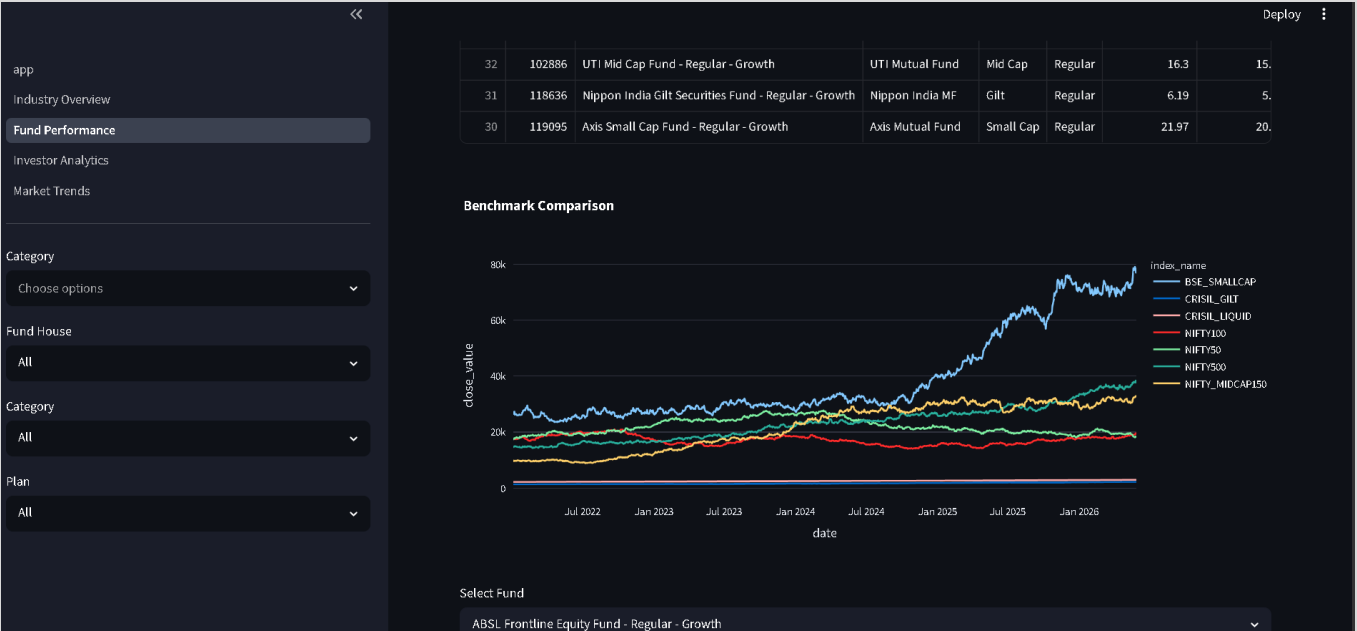
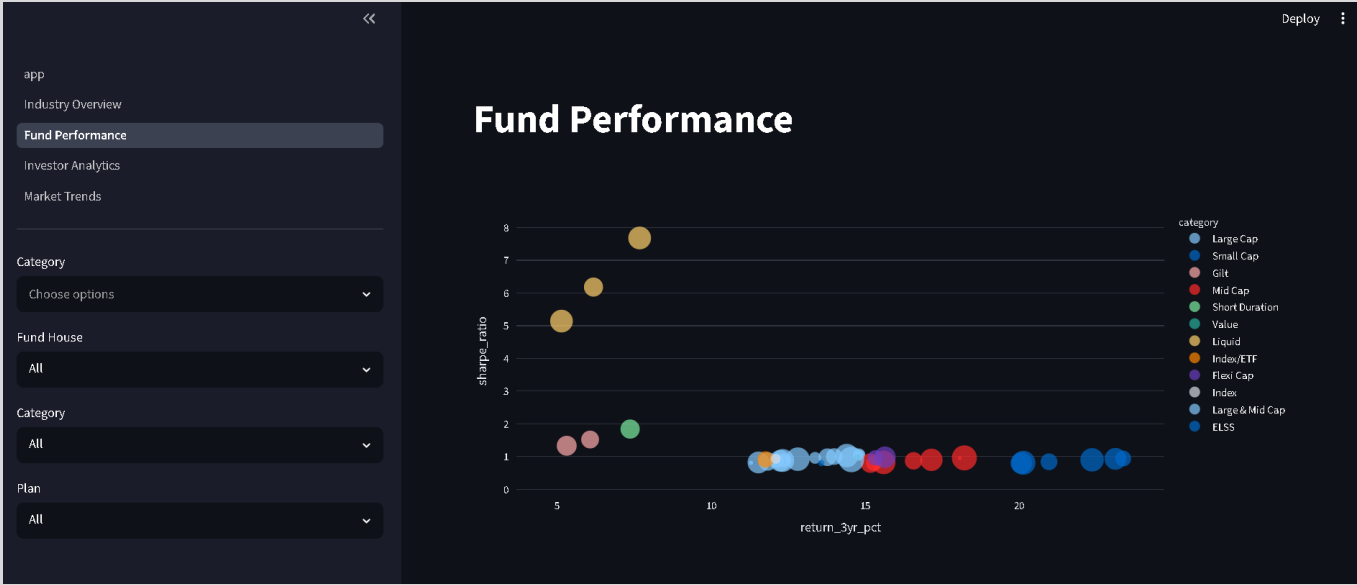
Expense Ratio

Findings

Large-cap funds displayed lower volatility.

Moderate-risk funds generated balanced returns.

High Sharpe Ratio funds demonstrated superior risk-adjusted performance.



DASHBOARD DEVELOPMENT

Streamlit Dashboard

An interactive dashboard was developed using Streamlit and Plotly.

Dashboard Modules

Home Page

Project overview and navigation.

Industry Overview

Industry-level KPIs and trends.

Fund Performance

Risk-return analytics and scorecards.

Investor Analytics

Investor demographics and transaction insights.

Market Trends

Benchmark and SIP trends.

app

Industry Overview

Fund Performance

Investor Analytics

Market Trends

Dashboard Navigation

Use the sidebar to explore:

• Industry Overview

• Fund Performance

• Investor Analytics

• Market Trends

Bluestock Internship Project



Bluestock Mutual Fund Analytics Dashboard

Analyze Mutual Fund Performance, Investor Behaviour, SIP Trends, Portfolio Allocations and Market Benchmarks

Welcome

This dashboard was developed as part of the **Bluestock Mutual Fund Analytics Internship Project**.

The platform provides insights into:



Mutual Fund Performance



SIP Growth Trends



Fund House Analysis



Investor Demographics



Market Benchmarks



Fund Scorecards

Quick Dashboard Overview

Performance Analytics

SIP Insights

Fund Houses

Market Benchmarks

app

Industry Overview

Fund Performance

Investor Analytics

Market Trends

Dashboard Navigation

Use the sidebar to explore:

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• Market Trends

Risk vs Return

Sharpe Ratio

Alpha/Beta

Fund Scorecards

SIP Trends

Category Inflows

Benchmark Analysis

Sector Allocation

Bluestock Mutual Fund Analytics

Interactive Dashboard for Mutual Fund Performance, Investor Analytics, SIP Trends and Market Insights

Built with Streamlit • Python • SQLite • Plotly

Bluestock Internship Program 2025

Designed & Developed by Akshat Bansal

[illegible]

app

Industry Overview

Fund Performance

Investor Analytics

Market Trends

State

Choose options

Fund House

All

Category

All

Plan

All

Investor Analytics

State	amount_inr
West Bengal	295M
Uttar Pradesh	285M
Telangana	290M
Tamil Nadu	315M
Rajasthan	298M
Punjab	318M
Maharashtra	270M
Madhya Pradesh	308M
Karnataka	275M
Haryana	280M
Gujarat	295M
Delhi	288M

LIMITATIONS

Limitations

- Simulated NAV values used for analysis.
- Limited historical period available.
- No real-time market integration.
- Dashboard operates on static snapshots.
- Advanced machine learning models not implemented.

RECOMMENDATIONS

Recommendations

- Integrate live AMFI APIs.
- Add portfolio optimization models.
- Implement machine learning recommendations.
- Develop cloud-hosted dashboard.
- Incorporate real-time benchmark tracking.

FUTURE ENHANCEMENTS

Future Scope

- Potential enhancements include:
- AI-powered recommendation engine.
- Portfolio simulator.
- Monte Carlo forecasting.
- Mobile application.
- Automated investor alerts.

CONCLUSION

Conclusion

The Mutual Fund Analytics Capstone Project successfully demonstrates an end-to-end analytics solution for mutual fund data.

The project combines ETL processes, exploratory analysis, advanced risk metrics, and interactive visualization to generate meaningful investment insights.

The developed dashboard provides a scalable foundation for future financial analytics applications and showcases practical applications of data science within the mutual fund industry.

REFERENCES

References

- [Pandas Documentation](#)
- [NumPy Documentation](#)
- [Streamlit Documentation](#)
- [Plotly Documentation](#)
- [SQLite Documentation](#)
- [AMFI India](#)
- [Bluestock Internship Resources](#)



THANK YOU!